

Industrial Security (Bachelor)

Solution Sheet – Section 02 ICS and SCADA Fundamentals

Exercise 2.1 – Component Mapping

1. Historian.
2. SIS (Safety Instrumented System).
3. PLC.
4. HMI.
5. RTU (legacy SCADA over leased serial / radio link).
6. DCS (Distributed Control System).

Exercise 2.2 – Sample Solution

```
| Start      Stop(NC)  Guard1  Guard2      Motor  |
|---] [---]/[-----] [-----] [----- (  ) ----|
|                                                    |
| Motor                                             |
|---] [-----+ (parallel seal-in into Motor)      |
|                                                    |
| Motor                      Lamp                  |
|---] [----- (  ) -----|
```

The seal-in latches the motor on; either Stop, Guard1, or Guard2 dropping out breaks the chain.

Exercise 2.3 – Independent SIS

1. Independence ensures that the BPCS cannot be a common-cause failure for the protection layer. If the same controller both runs the process and protects it, a single attack or fault can defeat both – giving an attacker direct path to unsafe states.
2. TRITON (2017) targeted a Schneider Triconex SIS at a Saudi petrochemical plant via the proprietary TriStation protocol. The goal was to reprogram safety logic to permit unsafe states. A bug in the malware caused the SIS to trip and revealed the operation.
3. (a) Signed and verified safety-logic uploads; (b) physical keyswitch for “program enable” that cannot be overridden remotely.

Exercise 2.4 – Lab Notes

Expected observations:

- OpenPLC defaults to Modbus TCP on port 502.
- HMI traffic uses function codes 0x03 (read holding regs), 0x06 or 0x10 (write).
- Communication is plaintext – no authentication, no integrity.